MAHATMA GANDHI UNIVERSITY
MG-UGP (HONOURS)REGULAR EXAMINATION MARCH 2026
FOURTH SEMESTER

DSC - MG4DSCMOS200 – WIRELESS COMMUNICATION TECHNOLOGY

(2024 ADMISSION ONWARDS)

Part A, Answer all questions, 1 mark each

1. **b** Different frequencies to different users
2. **b** FDMA
3. **b** USRP
4. **c** Better noise performance
5. **c** A unique spreading code
6. **c** Signal processing algorithms
7. **d** Microwave band
8. **b** Managing radio resources and handovers at the cell level
9. **c** Development cost and time
10. **c** HF, VHF, and UHF bands

Part B , Answer any 4 questions 5 marks each

11. (Explanation 3 Marks, limitations 2 marks)

Free-space wireless communication refers to the transmission of electromagnetic waves through air or vacuum without any guiding medium such as cables or fibers. The signal propagates directly from transmitter to receiver through line-of-sight or indirect paths.

Limitations:

- High path loss due to signal spreading
- Susceptibility to atmospheric conditions (rain, fog, etc.)
- Interference from other wireless signals
- Limited range without repeaters
- Multipath fading in urban environments

12. (5 layers – 1 mark each)

The Bluetooth protocol stack is a layered architecture used for short-range wireless communication.

It consists of:

- **Radio Layer:** Handles physical transmission of signals.
- **Baseband Layer:** Manages link establishment and packet handling.
- **Link Manager Protocol (LMP):** Controls authentication, encryption, and power modes.
- **L2CAP (Logical Link Control and Adaptation Protocol):** Provides multiplexing and segmentation.
- **Upper Layers:** Includes RFCOMM (serial communication), SDP (service discovery), and application profiles.

Bluetooth enables low-power, short-range communication between devices such as phones, laptops, and IoT devices.

13. (blocks – 2.5 marks, flowgraph – 2.5 marks)

In GNU Radio, signal processing is designed using graphical models.

- **Blocks:** These are basic processing units such as filters, modulators, demodulators, or signal sources/sinks.
- **Flowgraphs:** A flowgraph is a graphical representation of signal flow created by connecting blocks.

Each block performs a specific function, and data flows from source blocks to sink blocks through processing blocks. Flowgraphs can be built using GNU Radio Companion (GRC), making SDR design flexible and modular.

14. (Advantages 3 points – 2.5 marks, limitation 3 points – 2.5 marks)

Advantages:

- Wireless mobility and convenience
- Easy installation and scalability
- Supports multiple devices
- High data rates (especially modern standards like Wi-Fi 5/6)

Limitations:

- Limited range (especially indoors)
- Interference from other devices (microwave, Bluetooth)
- Security vulnerabilities if not properly configured
- Speed decreases with distance and obstacles

15. (Developments and explanation - 3 mark, Applications - 2 mark)

Recent developments:

- Integration with AI/ML for adaptive signal processing
- Use of high-performance FPGAs and GPUs
- Cloud-based SDR platforms
- Improved open-source frameworks like GNU Radio
- 5G and 6G research using SDR testbeds
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Applications:

- Military communication systems
- Cognitive radio networks
- Spectrum monitoring and analysis
- Emergency communication systems
- Research and prototyping of wireless systems

16. (Explanation – 5 marks)

The **Global Positioning System (GPS)** works based on satellite triangulation.

- A GPS receiver gets signals from at least four satellites.
- Each satellite sends its position and time information.
- The receiver calculates the distance using signal travel time.
- Using trilateration, the exact position (latitude, longitude, altitude) is determined.

GPS works globally and is widely used in navigation, mapping, and tracking applications.

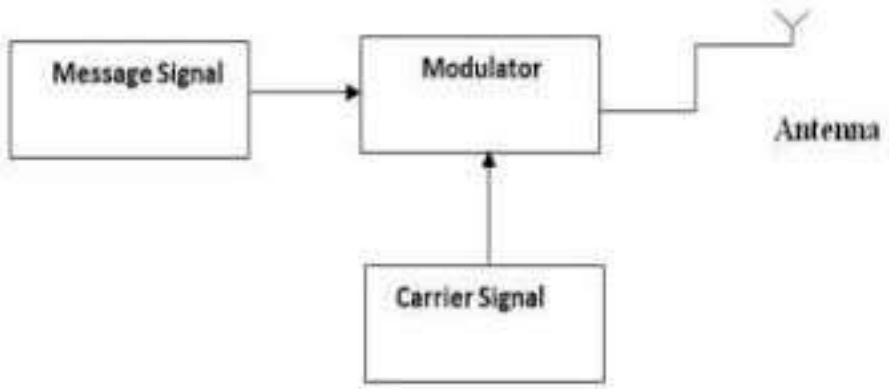
<u>Part C, Answer any 2 questions, 10 marks each</u>

17.(Basic principle, AM, FM, PM – 5 marks, need for modulation – 5 marks)

Modulation is the process of varying a high-frequency carrier signal in accordance with a low-frequency information (message) signal. It enables efficient transmission of signals over long distances in wireless communication.

A message signal (baseband) is superimposed on a carrier wave by varying its:

- Amplitude → AM (Amplitude Modulation)
- Frequency → FM (Frequency Modulation)
- Phase → PM (Phase Modulation)



1.Reduction of Antenna Height

Antenna height depends on wavelength:

$h=\lambda/4$

Since message signals have low frequency → large wavelength → impractical antenna size.

By using high-frequency carrier → antenna becomes practical.

2. Avoiding Interference
Without modulation, all baseband signals overlap.
Different carrier frequencies allow multiplexing and separation of signals.
3. Multiplexing Capability
Multiple signals can be transmitted simultaneously using different carriers (FDM).

18. (Need for modulation - 5 marks, noise and explanation – 5 marks)
Need for Modulation

- Reduces antenna size
- Prevents signal mixing
- Enables long-distance transmission
- Supports multiplexing

Types of Noise
1. Thermal Noise

- Caused by random motion of electrons
- Present in all electronic devices
- White noise in nature

2. Atmospheric Noise

- Due to lightning and natural disturbances
- Common in radio communication

3. Shot Noise

- Occurs in semiconductor devices
- Due to discrete nature of charge carriers

4. Interference Noise

- Caused by overlapping signals from other transmitters
- Man-made noise

Effect of Noise on Signal Transmission

- Reduces signal-to-noise ratio (SNR)
- Causes distortion
- Increases error rate
- Limits communication range

19. (objectives and philosophy – 5marks, Importance of FOSS – 5marks)
Objectives of GNU Philosophy

- Freedom to run software for any purpose
- Freedom to study and modify source code
- Freedom to redistribute copies
- Freedom to improve and share improvements

Philosophy of GNU

GNU emphasizes user freedom over proprietary control, promoting collaboration and transparency. The philosophy of GNU software is an ethical and social movement prioritizing user freedom and community collaboration over proprietary control. It asserts that software should respect fundamental user liberties, independent of its cost.

Importance of Free and Open Source Software (FOSS)

1. Cost Efficiency

- No licensing cost
- Suitable for academic and research systems

2. Transparency

- Source code is open and verifiable

3. Customization

- Can be modified for specific communication applications

4. Security

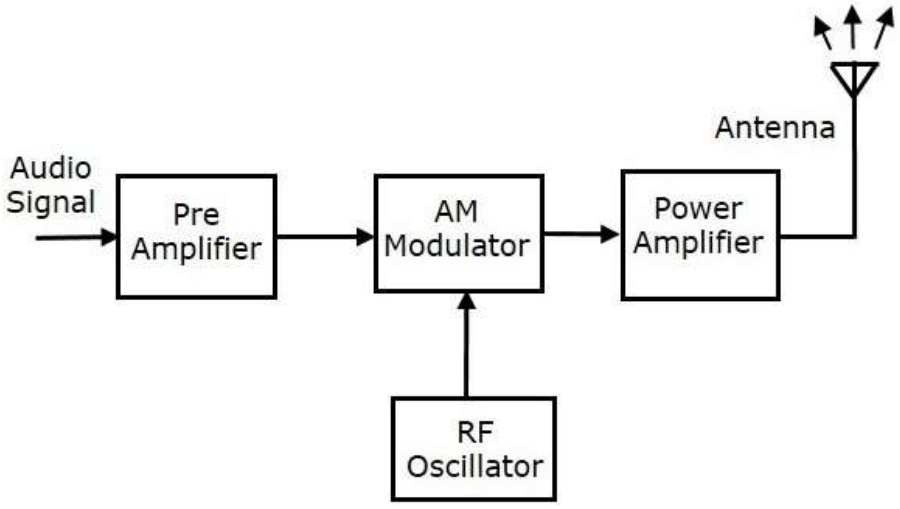
- Bugs are identified and fixed quickly by community

5. Innovation in Communication Systems

- Used in SDR platforms like GNU Radio

20. **(AM transmitter – block diagram -2marks, explanation -2marks, FM transmitter – block diagram – 2 marks, explanation – 2 marks, difference – 1 marks)**

AM transmitter

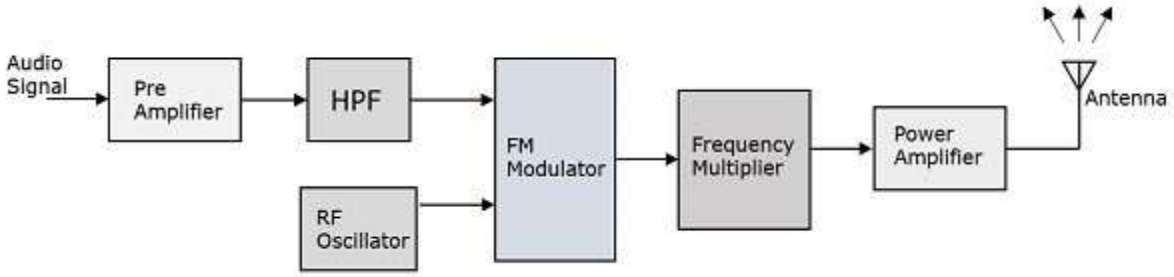


An Amplitude Modulation (AM) transmitter is a device that encodes audio or data onto a high-frequency carrier wave by varying the wave's amplitude. It operates in the medium frequency (MF) band (535 to 1705 kHz) and is used in broadcasting, aviation, and DIY radio testing.

Working

- Message signal is combined with carrier
- Amplitude of carrier varies
- Signal transmitted via antenna

FM transmitter



An FM (Frequency Modulation) transmitter takes an audio signal (like a voice or music) and encodes it onto a high-frequency radio wave for wireless transmission. The transmission process sequentially processes the audio signal, generates a carrier wave, combines them, and prepares the final signal for broadcast.

Key Differences

Feature	AM Transmitter	FM Transmitter
How it Works	Changes the wave's amplitude.	Changes the wave's frequency.
Audio Quality	Lower fidelity, limited frequency response.	High fidelity, crisp sound, excellent for music.
Signal Interference	Highly susceptible to static and electrical noise.	Highly resistant to static and noise.
Transmission Range	Long-range (can bounce off the ionosphere, especially at night).	Short-range (typically line-of-sight, 30–40 miles).
Bandwidth	Narrower, allowing for more stations.	Wider, requiring more spectrum space per station.